Practical 8

MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

1. Is the schema normalized properly?

Answer:

Yes, the StudentHub database is normalized by separating the data into three tables: students, events, and registrations. Each table stores specific information, reducing data duplication and maintaining data consistency. The registrations table connects students with events and helps implement a many-to-many relationship.

https://www.geeksforgeeks.org/dbms/introduction-of-database-normalization/

2. Are primary keys and foreign keys correctly defined?

Answer:

Yes, primary keys uniquely identify each record in their respective tables. The student_id in the students table and event_id in the events table are primary keys. The registrations table contains student_id and event_id as foreign keys referencing their respective tables. These constraints maintain referential integrity.

https://www.geeksforgeeks.org/dbms/difference-between-primary-key-and-foreign-key/

3.Is the database connection handled securely?

Answer:

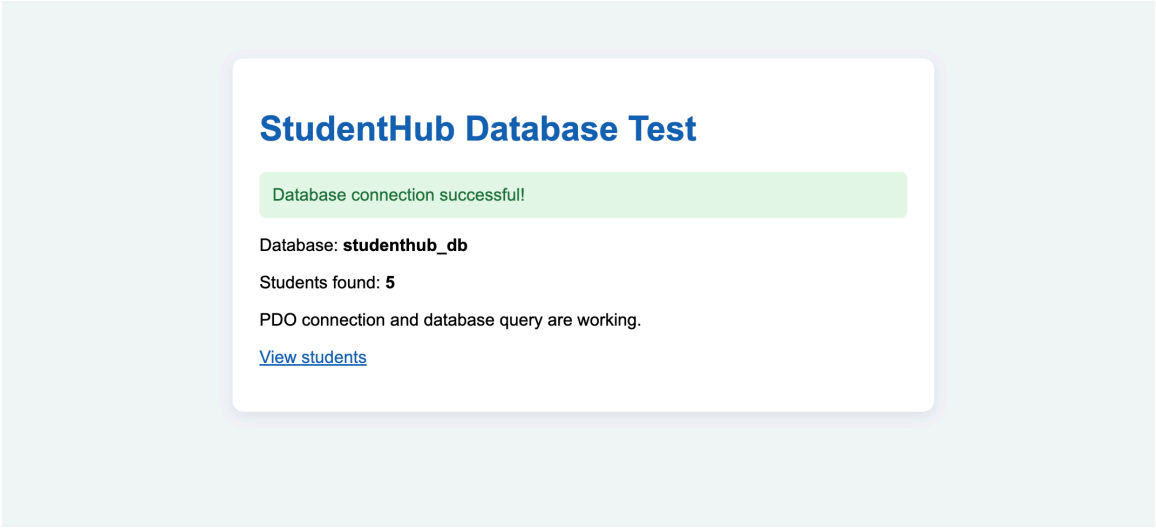
Yes, the database connection is handled securely using PHP PDO. Database credentials are kept in a separate db.php file, exceptions are used for error handling, and prepared statements are used to execute SQL queries safely. Prepared statements help prevent SQL injection attacks.

https://www.geeksforgeeks.org/php/php-data-objects-pdo/

B. Result

Thus, the StudentHub database was designed using MySQL with properly normalized tables, primary and foreign key constraints, and secure PHP PDO connectivity. Prepared statements help prevent SQL injection and ensure safe database operations.

C. Evidence of Work (Screenshots)



D. GitHub Repository Link:

https://github.com/Dhanraj2408/StudentHub

E. Challenges Faced:

Designing a normalized database structure for students, events, and registrations. Establishing relationships between tables using primary keys and foreign keys.Connecting PHP to MySQL using PDO.

F. Learning Outcomes:

Understanding how to design a normalized MySQL database for the StudentHub application, creating tables with primary and foreign keys, managing databases using phpMyAdmin, establishing a secure database connection using PHP PDO, and implementing prepared statements to prevent SQL injection and maintain data integrity.

Faculty Remarks: